

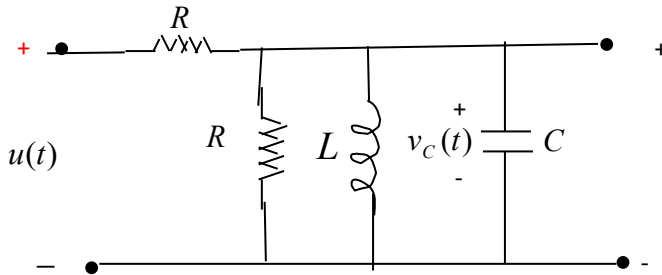
EEEE- 414: Classical Control

Homework Assignment # 2

Due date: by 11:59 PM, Thursday, September 10, 2026

Problem #1

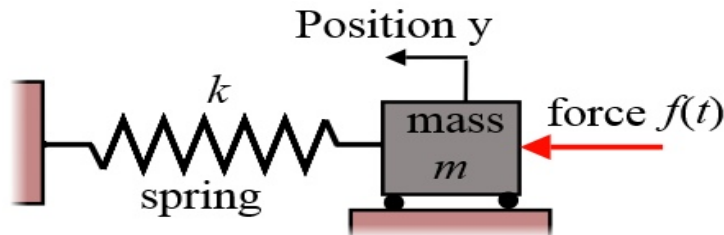
For the following RLC circuit. Assume $R = 10K\Omega$, $L = 200H$, and $C = 10\mu F$



Find the capacitor voltage for the unit step input. Plot the capacitor voltage (use MATLAB)

Problem # 2

Consider the mass-spring system shown below. Find $y(t)$ when the input is $f(t) = u(t)$



$$m = 2Kg$$

$$k = 6 N/m$$

$$v = \text{friction coefficient} = 2 Ns/m$$

Problem # 3

For each of the second order system shown below, find natural frequency, damping factor, rise time, settling time, time to peak, and OS.

a)

$$T(s) = \frac{16}{s^2 + 3s + 16}$$

b)

$$T(s) = \frac{0.04}{s^2 + 0.02s + 0.04}$$

c)

$$T(s) = \frac{1.05 \times 10^7}{s^2 + 1.6 \times 10^3 s + 1.05 \times 10^7}$$

Problem # 4

Derive an equation for the settling time of a second order system with damping factor of one (critically damped system).

Problem # 5

An autonomous system has an orientation system with transfer function

$$\frac{\theta(s)}{\theta_{ref}(s)} = \frac{53.176}{4.6s^2 + 31.281s + 53.176}$$

Make a plot of $\theta(t)$ in response to $\theta_{ref}(t) = 3u(t)$

Problem # 6

- a) Find the transfer function of a second-order system that yields a 15% overshoot and settling time of 0.7 second.
- b) Write an expression for the step response of the system.